



Semester One Examinations 2021 / 2022

Exam Code(s) 4BCT1, 4BSE1, 1MSME1, 1MEEE1, 1MECS1, 1CSD1, 1CSD2, 2SPE1, 1MAI1

Exam(s) B.Sc. Degree (Computer Science and Information Technology)
B.E. Energy Systems Engineering
MSc in Mechanical Engineering
ME (Electrical and Computer Engineering)
ME (Electrical and Electronic Engineering)
M.Sc. in Computer Science (Data Analytics)
M.Sc. in Computer Science (Artificial Intelligence)

Module Code(s) CT561
Module(s) Systems Modelling and Simulation

Paper No. I

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Instructions: Answer 3 questions out of 4. All questions carry equal marks.

Duration 2 hrs
No. of Pages 5 (Including Cover Page)
Department(s) School of Computer Science

1. (a) Describe how the following flows are formulated, and discuss how they can be used as building blocks for larger models.

- Fractional Increase Rate
- Fractional Decrease Rate

[4]

- (b) Construct a stock and flow model, showing link and loop polarity, from the following description for a University (it is a two-stock model)

- **Students** are increased by recruitments, and reduced by two flows: (1) graduations and (2) student withdrawals
- **Staff** are increased by new hires, and reduced by (1) resignations and (2) retirements
- The *student to staff ratio* is a ratio of Students (numerator) to Staff (denominator).
- The *desired staff level* is calculated from the number of students (numerator) and the *desired student to staff ratio* (denominator).
- New hires are based on a goal adjustment structure involving *desired staff level*, *staff*, *discrepancy* between goal and actual, and the *adjustment time*.
- As the *student to staff ratio* increases, the number of student withdrawals increases.

From this model, explain the following feedback loops and show clearly how their polarity is calculated. Discuss their importance for university planners.

- The hiring loop, by tracing from an increase in the student's stock, through to its impact on the staff stock, and back to the student stock again.
- The withdrawals loop, tracing from an increase in students to the withdrawals flow, and back to students again.

[15]

- (c) Formulate the staff management part of the model (new hires) as a stock management structure.

[6]

2. (a) Present the following differential equation model as a stock and flow model, with equations, where P is the city population, r is the growth rate, C is the carrying capacity, and s is the decline fraction.

Show the feedback loops, and the units of each equation in the stock and flow model.

Under what conditions would the value P reach the maximum value of C ?

$$\frac{dP}{dt} = r P \left(1 - \frac{P}{C} \right) - s P$$

[8]

- (b) For a vaccination rollout, two resources are required: (1) the vaccine and (2) a medical professional to administer the vaccine. Each medical person has a productivity, for example 20 vaccines/day.

Design a stock and flow model (with equations), that models the transition from Unvaccinated People to Vaccinated People. The flow controlling this transition (Vaccines Dispensed) is a *resource-based flow*, and, as such, it should capture the following logic.

Unvaccinated People	Number of Medics	Average Productivity	Total Daily Capacity	Number of Vaccines	Vaccines Dispensed
10,000	100	20	2,000	10,000	2,000
0	100	20	2,000	10,000	0
10,0000	0	20	0	10,000	0
10,0000	100	0	0	10,000	0
10,0000	100	20	2,000	10	10

[12]

- (c) A company's customer base can be modelled as a stock and flow system, with one inflow (customers gained) and one output (customers lost).

Show the stock and flow model (with equations). Assume that the fractional increase rate is 10%, and the fractional decrease rate is 3%.

Based on the net growth rate, predict how long it would take for the number of customers to double.

Confirm that this result is a good estimate, by assuming that the initial number of customers is 1,000. Make use of the following exact solution, where g is the net growth rate, C_t is the number of customers at time t , and C_0 is the initial number of customers.

$$C_t = C_0 e^{gt}$$

[5]

3. (a) Design an infectious disease model for a novel influenza pathogen with the following elements.
- People are initially *Susceptible*, and become *Exposed* at a rate λ (the force of infection). This overall force of infection is a summation of two different force of infections
 - The force of infection from infectious people.
 - The force of infection from quarantined people, assuming that quarantined people have 20% of the contacts of infectious people
 - People spend 2 days on average in the *Exposed* state, and this is a first order delay. People exit the *Exposed* state into an *Infectious* state, and remain there for 4 days, before moving to *Recovered*.
 - A fraction of people (governed by the quarantine fraction - QF) flow from the *Infectious* state into a *Quarantine* state. They remain infectious with reduced social contact once they are quarantined. Quarantined people remain infectious for 4 days, and then flow to *Recovered*.
 - Assume the average contact rate in the population is 10/day, and that infectivity is 0.1. The average contact rate for quarantined people is 2/day (20% of the contacts of those who are not quarantined).
 - Assume the total population is 100,000 and there are initially 10 people infected.

Based on this, develop:

1. A SEIR model, showing all feedbacks
2. A full set of equations, including the dimensions

[13]

- (b) Define the term reproduction number (R_0) and formulate an equation for R_0 in this model which takes into account the contribution from those who are infectious (and circulating with maximum contacts), and those who are infectious and quarantined (with fewer social contacts).

[7]

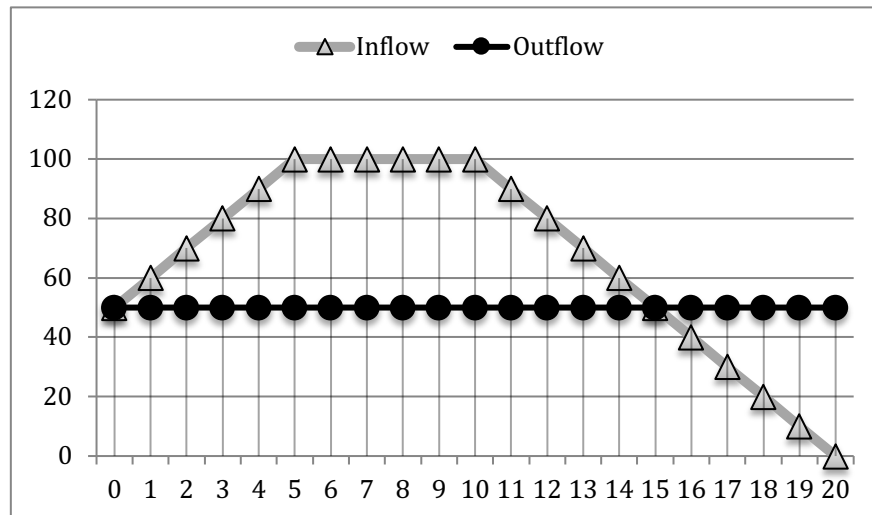
- (c) Estimate R_0 for the following range of scenarios

Scenario	Description	Parameters
1	No quarantine	QF=0
2	Some quarantine	QF=0.1
3	High quarantine	QF=0.5

[5]

4. (a) Based on the flow variables shown in the diagram, and assuming that the initial value of the stock is 100, perform the following:

- Draw a stock and flow model (including the net flow).
- Calculate, and show, the net flow.
- Use graphical integration to plot the stock over time, clearly explaining the different behaviour modes for each time interval
- Calculate the minimum, maximum and final value of the stock.



[12]

- (b) Describe Euler's method of integration, and state the underlying assumptions.

[5]

- (c) Calculate the stock using Euler's equation where $DT=5$. Estimate the integration error for each time step of 5.

How might you select an effective value for the time step?

[8]